Daniel Firoozabadi

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Education/Academic Summary

UC Berkeley - Electrical Engineering and Computer Engineering - GPA: 3.7/4.0
- **Relevant Coursework**: Introduction to Computer Programming for Scientists and Engineers; Structure and Interpretation of Computer Programs; Foundations of Signals, Dynamical Systems, and Information Processing; Multivariable Calculus
- **Organizations:** Undergraduate Real Estate Club, Berkeley Skydeck Fund, Berkeley Entrepreneurship Hub
- **Credit Hours:** 87.92, Expected Graduation: **Spring 2028**

Oak Park High School - GPA: 4.8/4.0 - Summa Cum Laude with Highest Honors
- **SAT**: 1550/1600
- **Organizations**: Math Club President, STEM Club President, NHS Treasurer, Science Olympiad team, FBLA, MHS Tutor

Leadership & Activities

Deloitte Salesforce Academy Bootcamp, 08/2025 to 08/2025
- One of 30 selected for an intensive program on Salesforce CRM, business process automation, and enterprise solution design.

Goldman Sachs Insight Series, 05/2025 to 07/2025
- Explored quantitative and engineering roles across divisions; participated in group challenges and market simulations

Susquehanna International Group Discovery Day, 01/2025 to 01/2025
- Explored quantitative trading, the role of global financial markets, and decision science in a competitive pre-internship program.

Point72 Academy Spring Sessions, 05/2025 to 06/2025
- Selected participant in Point72's selective educational program exploring hedge fund investing, equity research, and behavioral finance; learned from case-based workshops simulating the fundamental research process.

Cal Hacks X MTC Hackathon - 1st Place:
- Developed a web app leveraging Gemini's API to generate ethical score reports for companies, evaluating labor practices, climate impact, and human rights, and recommending more ethical alternatives.

Work/Research Experience

Software Engineering Intern - Olfera
05/2025 to 08/2025
- Built and deployed backend infrastructure using AWS Amplify, integrating authentication and CI/CD pipelines across dev and prod environments
- Designed and implemented scalable MySQL schemas to support platform entities (users, projects, experiments, SOPs) and enable fast, normalized querying
- Developed secure API endpoints with GraphQL/AppSync and contributed to React + TypeScript front-end development

Software Engineering Intern - Runopt
01/2025 to 05/2025
- Developed an advanced 3D visualization framework using PyVista for civil site design optimization
- Implemented a knapsack-based optimization algorithm to select from 9 UGDC models, improving storage-to-cost ratio within less than 5% of budget ceiling.
- Structured SQL tables for 20+ manufacturer specifications, enabling modular component selection across project types.
- Integrated OpenMDAO into the codebase, reducing cross-module optimization time by ~30%.

Student Researcher - UC Berkeley FHL Vive Center - Immersive Semi-Autonomous Aerial Command System (ISAACS) Research Group, 09/2024 to 12/24
- Developed and calibrated object detection algorithms using Zed SDK stereo cameras, increasing UAV mapping accuracy in real-time navigation tests.
- Improved depth estimation precision by ~15% through stereo vision enhancements and custom calibration routines.
- Contributed to flight software enabling UAVs to autonomously map environments with less than 0.5s processing latency.

Research Intern - University of Southern California, USC - Viterbi Flying Light Specks Lab
05/2023 to 08/2023
- Applied physics/engineering concepts in coding flight patterns for 5+ mini drones at a graduate-level lab.
- Programmed flight patterns for autonomous mini drones interfaced with Arduino by altering pitch, roll, and yaw
- Designated as a contributor to the NSF award of $250,000 for the project "EAGER: Prototyping Touch with FLS-Matter".

Technical Skills

Languages: Python, MATLAB, JavaScript, TypeScript, SQL, HTML/CSS
Frameworks/Tools: React, AWS Amplify, DynamoDB, AppSync, Pandas, NumPy, Node.js, OpenMDAO, PyVista, Excel
Technologies: GraphQL, Git, REST APIs, MySQL, COMSOL, Docker
Concepts: Optimization, Simulation, Numerical Methods, Signal Processing, Object-Oriented Design